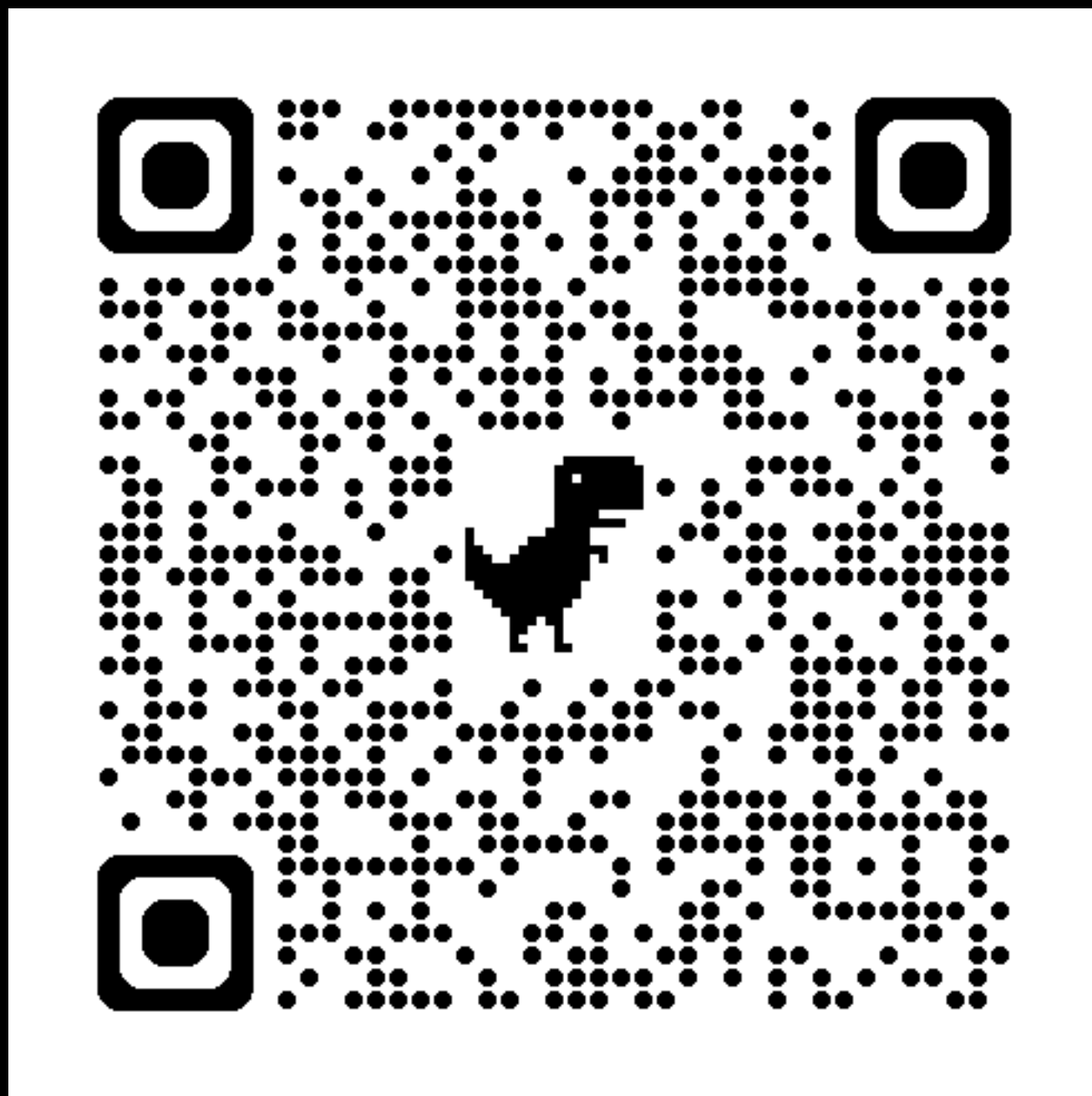


Interactive Demo: Gelbrich / Dual

1. Compute **exact discrete** W_2 using POT (linear program).
2. Extract **dual potentials** and verify feasibility: $f_i + g_j \leq C_{ij}$
3. Implement the **Gelbrich bound** and check that $\text{Bound} \leq \text{Exact}$



Hint for 1: check documentation for `ot.emd`

Hint for 2: broadcasting is your friend (but you can also use outer products)

For 3:

$$\|\mu_\alpha - \mu_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 \leq W_2^2(\alpha, \beta)$$

$$\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2 = \text{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2\left(\Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}\right)^{1/2}\right)$$

<https://bit.ly/cs2840-lecture4>

Interactive Demo: Gelbrich

